

with RHEL, to the extent that updates for RHEL can be used in CentOS and vice versa. If you're running CentOS, you're running RHEL.

There are a few key differences between RHEL and CentOS. Red Hat uses proprietary software for systems management, including the infamous *up2date*, which is used to download system updates from the Red Hat Update Network. CentOS replaces all the proprietary parts of the OS with open source components. Up2date is replaced with YUM (YellowDog Updater, Modified), the same tool used in Fedora, and CentOS uses YUM repositories for updates and software installation.

So what features does CentOS offer?

1. It's free of cost, and has community support. It's binary compatible with Red Hat Enterprise Linux.
2. It's rock solid — suitable for use in servers, clusters and workstations, as well as desktops and laptops.
3. It has integrated virtualisation support with a Xen kernel.
4. Included are tools for clustering and storage clustering.
5. The ability to run as just another desktop OS with all standard desktop features.

So let's get started with the review.

Test bed

A copy of Sun xVM VirtualBox 3.0.2 runs on Fedora 11's PAE kernel on an Intel Core2Duo E7200 processor (2.53 GHz and without VT). The VM has 1 GB of RAM.

First impressions

Before version 5.3, I'd used a customised version of CentOS 5.1. The first thing that struck me, from the installer itself, was the new Artwork stack. The CentOS developers have outdone themselves in this sphere, where it's normally accepted that all servers should look ugly. Speaking of installation, it's the same old Anaconda at version 11.1.2.168, with all the usual options. Check the installation box for details.

The desktop

In CentOS, you can choose between KDE 3.5.4 and GNOME 2.16.0. Though I prefer neither, for stability and familiarity's sake, I chose GNOME for the review. Visually, nothing has changed; it's the same old icons, the same old pointers and the same old GNOME. Something that could have been done was to add more shortcuts to administration tools. The new wallpaper, however, is refreshing.

The kernel version stands at 2.6.18-125.2.1.el5. Throughout a major version of CentOS and RHEL (such as 5.x), the versions of software cannot change. So, because CentOS 5.0 came out with KDE 3.5.4 (and kernel 2.6.18), all minor versions of CentOS 5 have to carry those versions of software. No cosmetic fixes are provided. Updates are necessary bug fixes or security fixes only. Therefore, OpenOffice stands at version 2.3, The GIMP at version 2.2, GCC at version 4.1.2, etc. So laugh out loud.

Installation

CentOS installs using Anaconda, one of the most modular Linux installers out there. It has a GUI mode as well as a text mode.

At the boot prompt, type in the following:

- `linux text vga=792` for a 1024x768 text mode installer
- `linux resolution=1024x768` for a 1024x768 GUI installer

The choice you make here reverberates throughout the installation. A text mode installation will disable automatic X start-up at system start, will not provide a graphical boot with RHGB, and all system config tools installed (including the firstboot manager) will be CLI-based. On the other hand, a GUI installation will make X run at start-up, install GUI versions of system config tools and will enable RHGB for a graphical boot.



Figure 1: Anaconda installer

The server within

Jokes apart, let's come in to what CentOS actually is — a server operating system. Right from installation, CentOS is optimised for server functions. The 'Recommended Partitioning Scheme' involves creating a separate /boot partition, and puts all the other partitions into an LVM. Pretty cool, because 'Dynamic Storage Scalability' (whatever that may be) is a big thing in servers.

At the Web stable, Apache version 2.2.3-22 heads the stack, with PHP at version 5.1.6. I have had bad experiences with that PHP version; it would not let me run RoundCube Webmail on it. MySQL and PostgreSQL are at versions 5.0.45 and 8.1.11. There you have it, your LAMP or LAPP stack. Personally, I've got nothing against this except the old PHP. PHP 5.2.10 and 5.3 SRPMs (Source RPMs) for CentOS are available at <http://www.cfsrpm.net/php/>. You'll need to build them. You'll also need to have the EPEL repositories added.

When it comes to FTP, VSFTPD 2.0.5 just doesn't cut it. It isn't as popular as ProFTPD, and people say it's not as easy to configure. I say, ditch it and install ProFTPD, which is available in the RPMForge repos.